

1. **KMS Corporation has assets with a market value of \$500 million, \$50 million of which are cash. It has debt of \$200 million, and 10 million shares outstanding. Assume perfect capital markets.**
 - a. What is its current stock price?
 - b. If KMS distributes \$50 million as a dividend, what will its share price be after the dividend is paid?
 - c. If instead, KMS distributes \$50 million as a share repurchase, what will its share price be once the shares are repurchased?
 - d. What will its new market debt-equity ratio be after either transaction?
 - a. $(500 - 200)/10 = 30$
 - b. $(450 - 200)/10 = 25$
 - c. $(450 - 200)/(10 - 1.667) = 30$
 - d. $200/250 = 0.8$

2. **Natsam Corporation has \$250 million of excess cash. The firm has no debt and 500 million shares outstanding, with a current market price of \$15 per share. Natsam's board has decided to pay out this cash as a one-time dividend.**
 - a. What is the ex-dividend price of a share in a perfect capital market?
 - b. If the board instead decided to use the cash to do a one-time share repurchase, in a perfect capital market what is the price of the shares once the repurchase is complete?
 - c. In a perfect capital market, which policy, in part (a) or (b), makes investors in the firm better off?
 - a. The dividend payoff is $\$250/500 = \0.50 on a per share basis. In a perfect capital market the price of the shares will drop by this amount to \$14.50.
 - b. \$15
 - c. Both are the same.

3. **Suppose the board of Natsam Corporation decided to do the share repurchase in Problem 7 part (b), but you, as an investor, would have preferred to receive a dividend payment. How can you leave yourself in the same position as if the board had elected to make the dividend payment instead?**

If you sell 0.5/15 of one share you receive \$0.50 and your remaining shares will be worth \$14.50, leaving you in the same position as if the firm had paid a dividend.

4. **Suppose you work for Oracle Corporation, and part of your compensation takes the form of stock options. The value of the stock option is equal to the difference between Oracle's stock price and an exercise price of \$10 per share at the time that you exercise the option. As an option holder, would you prefer that Oracle use dividends or share repurchases to pay out cash to shareholders? Explain.**

Because the payoff of the option depends upon Oracle's future stock price, you would prefer that Oracle use share repurchases, as it avoids the price drop that occurs when the stock price goes ex-dividend.

5. **Suppose B&E Press paid dividends at the end of each year according to the schedule below. It also reduced its share count by repurchasing 5 million shares at the end of each year at the ex-dividend stock prices shown. (Assume perfect capital markets.)**

	2009	2010	2011	2012	2013
Ex-Dividend Stock Price (\$/share)	10.00	12.00	8.00	11.00	15.00
Dividend (\$/share)		0.50	0.50	0.50	0.50

Shares Outstanding (millions)	100	95	90	85	80
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- What is total market value of B&E's equity, and what is the total amount paid out to shareholders, at the end of each year?
- If B&E had made the same total payouts using dividends only (and so kept its share count constant), what dividend would it have paid and what would its ex-dividend share price have been each year?
- If B&E had made the same total payouts using repurchases only (and so paid no dividends), what share count would it have had and what would its share price have been each year?
- Consider a shareholder who owns 10 shares of B&E initially, does not sell any shares, and reinvests all dividends at the ex-dividend share price. Would this shareholder have preferred the payout policy in (b), (c), or the original policy?

	2009	2010	2011	2012	2013	
a. Ending Equity Value	1000	1140	720	935	1200	Equity Value = Stock Price × Shares Outstanding
Total Dividends		50.00	47.50	45.00	42.50	Total Dividends = Dividend × Shares
Total Repurchases		60.00	40.00	55.00	75.00	Total Repurchases = Shares Repurchased × Share Price
Total Payout		110.00	87.50	100.00	117.50	Total Payout = Total Dividends + Repurchases
b. Dividends Only						
Shares	100	100	100	100	100	Shares = Initial Shares
Dividend (\$/share)		1.10	0.88	1.00	1.18	Dividend = Total Payout / Initial Shares
Stock Price (ex-div)	10.00	11.40	7.20	9.35	12.00	Stock Price = Equity Value / Initial Shares
c. Repurchases Only						
Beginning Shares		100.00	91.20	81.32	73.46	Beginning Shares = Last Period's Ending Shares
Stock Price	10.00	12.50	8.85	12.73	17.93	Stock Price = (Equity Value + Total Payout) / (Beginning Shares) ¹
Shares Repurchased		8.80	9.88	7.86	6.55	Shares Repurchased = Total Payout / Stock Price

¹ because stock price is the same before and after repurchase

Ending Shares	100	91.20	81.32	73.46	66.91	Ending Shares = Beginning Shares – Repurchased Shares
Ending Equity Value	1000	1140	720	935	1200	Check to see that this matches our original results.

d. The Modigliani-Miller Dividend Irrelevance proposition implies that the shareholder should be indifferent. As long as the investor keeps the fraction invested the same (in this case, fully invested), the value of her portfolio should be independent of the policy.

We can check this by noting the total return on the stock each year is the same in all cases. Alternatively, we can explicitly track the portfolio value.

Investor Holdings (assuming full reinvestment)

	2009	2010	2011	2012	2013
Original Policy					
Stock Price	10.00	12.00	8.00	11.00	15.00
Dividend		0.50	0.50	0.50	0.50
<i>Total Return</i>		25.0%	–29.2%	43.8%	40.9%
Initial Shares		10.00	10.42	11.07	11.57
Dividend Received		5.00	5.21	5.53	5.79
New Shares Purchased		0.42	0.65	0.50	0.39
Final Shares	10.00	10.42	11.07	11.57	11.96
Final Portfolio Value	100.00	125.00	88.54	127.28	179.35
Dividends Only					
Stock Price	10.00	11.40	7.20	9.35	12.00
Dividend		1.10	0.88	1.00	1.18
<i>Total Return</i>		25.0%	–29.2%	43.8%	40.9%
Initial Shares		10.00	10.96	12.30	13.61
Dividend Received		11.00	9.59	12.30	15.99
New Shares Purchased		0.96	1.33	1.32	1.33
Final Shares	10.00	10.96	12.30	13.61	14.95
Final Portfolio Value	0.00	125.00	88.54	127.28	179.35
Repurchases Only					

Stock Price	10.00	12.50	8.85	12.73	17.93
Dividend		0.00	0.00	0.00	0.00
Total Return		25.0%	-29.2%	43.8%	40.9%
Initial Shares		10.00	10.00	10.00	10.00
Dividend Received		0.00	0.00	0.00	0.00
New shares Purchased		0.00	0.00	0.00	0.00
Final Shares	10.00	10.00	10.00	10.00	10.00
Final Portfolio Value	100.00	125.00	88.54	127.28	179.35
Stock Price	10.00	12.50	8.85	12.73	17.93
Shares Repurchased		8.80	9.88	7.86	6.55
Shares	100	91.20	81.32	73.46	66.91
Equity Value	1000	1140	720	935	1200
Return		25.00%	-29.17%	43.75%	40.91%
Check		25.00%	-29.17%	43.75%	40.91%

6. Harris Corporation has \$250 million in cash, and 100 million shares outstanding. Suppose the corporate tax rate is 35%, and investors pay no taxes on dividends, capital gains, or interest income. Investors had expected Harris to pay out the \$250 million through a share repurchase. Suppose instead that Harris announces it will permanently retain the cash, and use the interest on the cash to pay a regular dividend. If there are no other benefits of retaining the cash, how will Harris' stock price change upon this announcement?

Effective tax disadvantage of retention is $t^* = 35\%$. (The reason is that Harris will pay 35% tax on the interest income it earns.) Thus, stock price falls by $35\% \times \$250\text{m}/100\text{m shares} = \0.875 per share.

7. Suppose that all capital gains are taxed at a 25% rate, and that the dividend tax rate is 50%. Arbuckle Corp. is currently trading for \$30, and is about to pay a \$6 special dividend.

- a. Absent any other trading frictions or news, what will its share price be just after the dividend is paid?

Suppose Arbuckle made a surprise announcement that it would do a share repurchase rather than pay a special dividend.

- b. What net tax savings per share for an investor would result from this decision?

- c. What would happen to Arbuckle's stock price upon the announcement of this change?

a. $t^*_d = (50\% - 25\%)/(1 - 25\%) = 33.3\%$, $P_{\text{ex}} = 30 - 6(1 - t^*) = \26

- b. With dividend, tax would be $6 \times 50\% = \$3$ for dividend, with a tax savings of $4 \times 25\% = \$1$ for capital loss, for a net tax from the dividend of \$2 per share. This amount would be saved if Arbuckle does a share repurchase instead.

- c. Stock price rises to by \$2 to \$32 to reflect the tax savings.

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